INCH-POUND

MIL-C-63227A (AR)
AMENDMENT 3
12 JULY 1989
SUPERSEDING
AMENDMENT 2

MILITARY SPECIFICATION

CONTAINER, AMMUNITION, METAL, PA 101 FOR
PROJECTILE, 155MM: HE, GUIDED, CANNON
LAUNCHED M712 AND TRAINING M823

This amendment forms a part of Military Specification MIL-C-63227A, dated 9 September 1981, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1 Delete the phrase "Clam Shell type container used for packing" and substitute the phrase "Clam Shell type container, Pallet, Dunnage, Bumpers and Filler Assembly used for packing and unitizing".

TITLE: Delete the phrase "155MM, HE, Guided, Cannon Launched" and substitute the phrase "155MM: Heat, Cannon Launched, Guided".

Delete the phrase "155MM, HE, Guided, Cannon Launched" and substitute the phrase "155MM: Heat, Cannon Launched, Guided".

PAGE 2

Delete paragraph 2.1 in its entirety and substitute the following:

"2.1 Government documents

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and standards (DoDISS) and supplement thereto, cited in the solicitation.

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SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-810 - Environmental Test Methods

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

9300440 - Container, Ammunition, Metal PA101, for Projectile, 155mm, Heat, Cannon Launched Guided, M712, and Training, M823.

9305335 - Packaging Assembly for Projectile, 155mm: Heat, Cannon Launched Guided, M712, and Training, M823 with packing and marking instructions

U.S. ARMY DEFENSE AMMUNITION CENTER AND SCHOOL

19-48-4159-20PM-1003 - Unitization Procedures for 155MM Copperhead seperate Loading Projectiles, Packed one per metal container, unitized 6 containers per special 4-way entry pallet.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951-82 - Standard Practice for Commercial Packaging"

Delete "FSC: 1320" and substitute "FSC 8140" at the right hand corner at the bottom of the page.

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* Delete the phrase "9305335 - Packaging Assembly for Projectile, 155MM: Heat" and substitute "9305335 - Packaging Assembly for Projectile, 155MM: HEAT".

2.1 Drawings. Delete the following:

"US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9300440 - Container, Ammunition, Metal PA101, for Projectile, 155MM, HE, Guided Cannon Launched, M712 and Training M823"

and substitute:

"U.S. ARMY ARMAMENT RESEARCH DEVELOPMENT AND ENGINEERING CENTER

9300440 - Container, Ammunition, Metal PA101, for Projectile, 155MM, HEAT, Cannon Launched Guided, M712 and Training M823"

Add the following:

"9305335 - Packaging Assembly for Projectile, 155MM: Heat, Cannon Launched Guided, M712, and Training, M823 with Packing and Marking Instructions

U.S. ARMY DEFENSE AMMUNITION CENTER AND SCHOOL

19-48-4159-20PM-1003 - Unitization Procedures for 155MM Copperhead Separate Loading Projectiles, Packed One Per Metal Container, Unitized 6 Containers Per Special 4-Way Entry Pallet"

* 3.2 Delete in its entirety and substitute the following:

"3.2 Components and assemblies.

3.2.1 Containers. The containers (container assemblies) shall comply with all requirements specified on drawing 9300440 and associated drawings and with all requirements specified in applicable specifications and standards.

3.2.2 Wooden assemblies. All wooden parts and assemblies used in unitization shall comply with all requirements specified on drawing 19-48-4159-20PM-1003 and associated drawings and with all requirements specified in applicable specifications and standards."

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3.4 Delete in its entirety and substitute the following:

"3.4 Environmental conditions. The container shall be laden with one Projectile, Training, M823 according to drawing 9305335. The Dummy Card, Label, and stenciling may be omitted. The Bands, Adjustment Handle, and Lifting Straps supplied inside the container shall be used as the internal packaging. Any prior damage exhibited by the projectile shall be recorded by markings, photographs, or documented list. The laden container shall withstand exposure to the tests and inspections as specified in Table I - First Article Inspection or Paragraph 4.4.2.27 of Quality Conformance Inspection - whichever is applicable.

The sequence of tests and inspections shall occur in the order as listed. No adjustments shall be made to the barrel nuts on the latches throughout the entire test and inspection sequence.

Examination shall commence with the unpacking of the projectile. The container lid shall be unlatched and opened without the use of a bar or any other device. There shall be no damage to or displacement of the gasket. The tail stop should be backed off without the use of any tool, except the adjustment handle. The projectile shall be manually removed from the container using lifting straps, and without prying.

There shall be no weld cracks visible except those allowable on containers which have been subjected to fragility shock exposure. There shall be no physical damage to the projectile or any one of the elements of the container assembly which would render them unserviceable as determined by the procuring agency. Minor dents, scratches and other forms of permanent deformations to the container that do not obstruct handling, assembly or ability of the container to protect its contents through its continued use shall not constitute cause for rejection.

Examination shall be complete when the projectile is repacked, as above, in the tested container.

The Government shall select the sample for mid test examination."

* 3.7 Delete in its entirety and substitute the following:

"3.7 Workmanship.

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3.7.1 Containers. The containers shall be regular, smooth, and free from wrinkles, pin holes, cracks, rough spots, burrs, sharp edges, and other defects that might affect the serviceability, durability, safety, and appearance of the container.

3.7.2 Wooden assemblies. Wooden assemblies shall be free of unrepaired splits, excessive knots or frayed edges which would impair the shipping and handling of the pallet unitized load. Units shall be sound. Straps shall be aligned and tensioned. Nails shall be driven flush or slightly below."

PAGE 4

Delete Table 1, First Article Inspection, and substitute new Table 1 included.

PAGE 9

* 4.4.2.4, Major Category: Add a new major defect as follows:

"Major 105 Flashing on inside surface 0.40% 3.2 Visual"

4.4.2.4, Major 103: Delete "Samll" and substitute "Small".

PAGE 12

4.4.2.8, Major Category: Add a new major defect as follows:

"109 Defective threads (like sharp crest of threads, burrs, etc.) 0.40% 3.2 Visual"

PAGE 13

4.4.2.9, Major Category: Add a new major defect as follows:

Major 107	Defective threads (like sharp crest of threads, burrs, etc.)	0.40%	3.2	Visual

PAGE 16

4.4.2.12: Delete the title "Pin, Latch" and substitute "Pin, Cage".

Major 102: Delete "Length of Small ID" and substitute "Length of Small OD".

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Assemblies and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Container Assemblies (Dwg. 9300440)				
	a. Examination for defects	4	100%	3.2	4.4.2.6
	b. Handle pull test	1		3.3	4.5.3
	c. Salt fog test	4	100%	3.4.1	4.5.1.1
	d. Transportation vibration test	4	100%	3.4.2	4.5.1.2
	e. Loose cargo vibration test	4	100%	3.4.3	4.5.1.3
	f. Transit drop test	4	100%	3.4.4	4.5.1.4
	g. Leakage test 1/	1		3.2	4.5.2
	h. Mid test examinations 1/	1		3.4	Visual
	i. Leakage test 1/	1		3.2	4.5.2
	j. Fragility shock test	4	100%	3.4.5	4.5.1.5
	k. Leakage test	4	100%	3.2	4.5.2
	l. Post test examinations 2/	4	100%	3.4	Visual
	Detailed Parts (Dwg. - See Para 4.4.2.1-4.4.2.25				
	Examination for defects	5 ea	100%	3.2	4.4.2.1 thru 4.4.2.25

NOTES: 1/ Same one sample unit shall be subjected to the tests and examinations indicated by g, h, and i above.

2/ Crack criteria for welding. After completion of fragility shock and leakage tests, the following criteria shall apply:

	On Each Side of Bulkhead	On Each TIG Weld	Around Each Cover Stop
Maximum number of cracks	2	1	1
Shortest distance in inches between two nearest cracks allowable	4	---	---
Maximum length of individual crack	0.5	0.25	0.25

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PAGE 17

4.4.2.13, Minor Category: Add a new defect as follows:

"202	Groove pin properly centered within the drawbolt	1.0%	3.2	Visual"

PAGE 25

4.4.2.21, Delete Page 25 for "Tab, Pre-load", and substitute new Page 25 for "Band, Preload" included.

PAGE 27

4.4.2.23, Major 106 Material Properties: Delete "0.40%" under heading AQL.

PAGE 28

4.4.2.24, Major Category:

Major 102: Delete the item in its entirety.

Major 103: Delete the item and substitute the following:

"103 Location of cover stops (at 6 places)"

PAGE 29

4.4.2.25, Major Category: Add major defects as follows:

"107	Anti-seize compound not clearly visible over entire threaded surface of aft bulkhead and tail stop	0.40%	3.2	Visual
108	Absence of properly applied anti-seize compound on both tail stop and aft bulkhead	0.40%	3.2	Visual
109	Presence of pain on threaded surface of both tail stop and aft bulkhead	0.40%	3.2	Visual

PAGE 30

4.4.2.26, Major 104: Delete the item in its entirety.

PAGE 30-A

4.4.2.27: Delete Page 30A added by Amendment 1, and substitute a new Page 30A included.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9301827
4.4.2.21	Band, Preload		NEXT HIGHER ASSEMBLY 9300440

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tab overall height		0.40%	3.2	Gage
102	Tab overall width		0.40%	3.2	Gage
103	Friction between tabs and band	Note a	0.40%	3.2	Manual
104	Location of tabs		0.40%	3.2	Gage
105	Strength requirements		0.40%	Dwg	Test
106	Material properties		0.40%	3.2	Certificate of Conformance
Minor					
201	Tab overall length		0.65%	3.2	Gage
202	Drawbolt thread engagement throughout entire length of torque nut		0.65%	3.2	Visual
203.	Evidence of poor workmanship		1.0%	3.7	Visual

NOTES:

a. Lack of friction shall be cause for rejection.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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PAGE 30-B

* Add Classification of Defects and Tests as shown in the included sheet (paragraph 4.4.2.28).

PAGE 30-C

* Add Classification of Defects and Tests as shown in the included sheet (paragraph 4.4.2.29).

PAGE 30-D

* Add Classification of Defects and Tests as shown in the included sheet (paragraph 4.4.2.30).

PAGE 30-E

* Add Classification of Defects and Tests as shown in the included sheet (paragraph 4.4.2.31).

PAGE 30-F

* Add Classification of Defects and Tests as shown in the included sheet (paragraph 4.4.2.32).

PAGE 31

4.4.3, Change paragraph number from "4.4.2.26" to "4.4.2.32".

5.4.1.4, Change paragraph number from "5.4.1.4" to "4.5.1.4".

5.4.1.5, Change paragraph number from "5.4.1.5" to "4.5.1.5".

PAGE 32

* 5.2 Delete in its entirety and substitute the following:

"5.2 Packing. Containers shall be unitized, 6 per pallet, in accordance with dwg. 19-48-4159-20PM-1003. Commercial strapping may be used in lieu of strapping specified."

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	Container Assembly	1 OF 1	9300440
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	a. Transportation vibration test	Note a	Note b	3.4.2	4.5.1.2
102	b. Loose cargo vibration test	Note a	Note b	3.4.3	4.5.1.3
103	c. Transit drop test	Note a	Note b	3.4.4	4.5.1.4
104	d. Handle pull test	Note c	Note b	3.3	4.5.3
105	e. Leakage test	Note a	Note b	3.2	4.5.2
106	f. Post test examinations	Note a	Note b	3.4	3.4
Minor	None defined				

NOTES: a. For lot size 2 to 50 - 2 samples; for lot size 51 to 100 - 3 samples; for lot size 501 to 3200 - 5 samples.
b. There shall not be any failure in any of the samples.
c. One sample in five lots.

AMSMC Form 1570a, 1 Apr 85 Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 19-48-4159-20PM-1003
4.4.2.28	Pallet Unit				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Field check		.40	3.2	Gage
102	Improper lumber group		1.0	3.2	Visual
103	Improper nails		1.0	3.2	Visual
104	Improper staples, seals or straps		1.0	3.2	Visual
105	No tension on straps		1.0	3.2	Manual
106	Container strapped over filler		100%	3.2	Visual
Minor					
201	Lifting handles inboard		1.0	3.2	Visual
202	Rope handles not retracted		1.0	3.2	Visual
203	Overall height, width, and depth		1.0	3.2	Gage
204	Workmanship		1.0	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.29	Special Pallet/Pallet Dunnage Location Detail		19-48-4159-20PM-1003
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Improper lumber group		1.0	3.2	Visual
102	Improper nails		1.0	3.2	Visual
103	Location of Dunnages A and B		1.0	3.2	Gage
104	Stop piece dimensions on A and B designated (ACTUAL)		1.0	3.2	Gage
Minor					
201	Special pallet heigth, width and length		1.0	3.2	Gage
202	Workmanship		1.0	3.7	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 19-48-4159-20PM-1003
4.4.2.30	Dunnages C, D, E and F		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Improper lumber group		1.0	3.2	Visual
102	Improper nails		1.0	3.2	Visual
103	Cross piece diemensions on E and F designated (ACTUAL)		1.0	3.2	Gage
Minor					
201	Overall width and length		1.0	3.2	Gage
202	Workmanship		1.0	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 19-48-4159-20PM-1003
4.4.2.31	Filler Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Location of filler pieces		1.0	3.2	Gage
202	Workmanship		1.0	3.7	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.32	L-Shaped Bumpers	1 OF 1	19-48-4159-20PM-1003
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Improper lumber group		1.0	3.2	Visual
102	Improper nails		1.0	3.2	Visual
Minor					
201	Overall height, width and length		1.0	3.2	Gage
202	Workmanship		1.0	3.7	Visual

NOTES:

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5.3 Delete in its entirety and substitute the following:

"5.3 Marking. Shipping marking shall comply with ASTM D 3951-82 and the following instructions. (No shipping marking shall be tagged "COPPERHEAD PACKING". Commercial shipping tags attached with wire shall be used unless otherwise specified. The following information shall be shown: Item name and model number, part number and revision, contractor number, lot number and date, name and address of manufacturer. If containers are bundled, boxed or palletized, the gross weight and quantity shall be included. The above information may be placed on pallets or boxes by labeling or stenciling. On the unitized assembly, each piece of Dunnage, C,D,E, and F shall be marked "C", "D", "E", and "F" respectively in letter size approximately one inch high. Black crayon or stencil is permissible.

5.4 Delete in its entirety.

PAGE 33

Delete "Project no. 1320" and substitute "Project no. 8140".

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army-AR

Preparing activity:
Army-AR

(Project 8140-A732)

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